12TH BIOLOGY · CHAPTER 19 · FULL CHAPTER MCQS

Chromosome and DNA

105 topic-wise MCQs from the complete chapter with answers and reasons

19.1 DETAILED STRUCTURE OF CHROMOSOME

1. Chromosomes were discovered in 1882 by:

(a) Waldeyer-Hartz (b) Johannsen
(c) Walther Flemming (d) Griffith

Answer: (c) Walther Flemming — He saw them in dividing cells of salamander larvae.

2. The complete set of chromosomes of a species is called its:

(a) Karyotype (b) Genome map
(c) Chromatin (d) Replicon

Answer: (a) Karyotype — It shows the number and shape of chromosomes.

3. The uncoiled network of chromosomes in a non-dividing cell is called:

(a) Chromatid (b) Chromatin
(c) Kinetochore (d) Satellite

Answer: (b) Chromatin — It is seen during interphase.

4. Sister chromatids are attached to each other at the:

(a) Telomere (b) Locus
(c) Satellite (d) Centromere

Answer: (d) Centromere — It is a constricted region.

5. The centromere is a specific DNA sequence of about:

(a) 80 nucleotides (b) 146 nucleotides
(c) 220 nucleotides (d) 1000 nucleotides

Answer: (c) 220 nucleotides — As stated in the textbook.

6. The disc-shaped protein structure where spindle fibres attach is the:

(a) Centromere (b) Kinetochore
(c) Telomere (d) Nucleosome

Answer: (b) Kinetochore — It is attached to the centromere.

7. The secondary constriction of a chromosome makes:

(a) Nucleoli (b) Spindle fibres
(c) Ribosomal subunits in cytoplasm (d) Telomeres

Answer: (a) Nucleoli — It is also called the organizer region (NOR).

studysapiens.com

8. The terminal ends of chromosomes are called:

(a) Satellites (b) Kinetochores
(c) Centromeres (d) Telomeres

Answer: (d) Telomeres — They stop chromosomes from joining end to end.

9. A chromosome with the centromere at its centre is:

(a) Acrocentric (b) Metacentric
(c) Telocentric (d) Sub-metacentric

Answer: (b) Metacentric — Its two arms are equal.

10. A chromosome with the centromere at the end is:

(a) Metacentric (b) Sub-metacentric
(c) Acrocentric (d) Telocentric

Answer: (d) Telocentric — Acrocentric has it near the end.

11. A chromosome is chemically made up of:

(a) 40% DNA and 60% proteins (b) 60% DNA and 40% proteins
(c) 50% DNA and 50% proteins (d) 90% DNA and 10% proteins

Answer: (a) 40% DNA and 60% proteins — The proteins are histones.

12. Histones are positively charged because they are rich in:

(a) Glutamic acid and valine (b) Methionine and serine
(c) Arginine and lysine (d) Phenylalanine and tyrosine

Answer: (c) Arginine and lysine — These are basic amino acids.

13. A histone octamer does NOT contain:

(a) H1 (b) H2A
(c) H3 (d) H4

Answer: (a) H1 — H1 is the linker histone outside the octamer.

14. The DNA wrapped around one histone octamer is about:

(a) 80 nucleotides (b) 250 nucleotides
(c) 220 nucleotides (d) 146 nucleotides

Answer: (d) 146 nucleotides — It wraps twice around the octamer.

15. The thickness of the nucleosome string is:

(a) 2 nm (b) 11 nm
(c) 30 nm (d) 300 nm

Answer: (b) 11 nm — The DNA helix itself is 2 nm.

16. The thickness of the chromatin fibre is:

(a) 11 nm (b) 300 nm
(c) 30 nm (d) 700 nm

Answer: (c) 30 nm — It is made of tightly packed nucleosomes.

studysapiens.com

17. The thickness of a chromatid is:

(a) 700 nm (b) 30 nm
(c) 300 nm (d) 11 nm

Answer: (a) 700 nm — The supercoil is 300 nm.

18. The inactive, dark-banded portion of chromatin is:

(a) Euchromatin (b) Heterochromatin
(c) Nucleosome (d) Linker DNA

Answer: (b) Heterochromatin — Euchromatin is the active portion.

19. The term “gene” was introduced in 1909 by:

(a) Mendel (b) Flemming
(c) Francis Crick (d) Wilhelm Johannsen

Answer: (d) Wilhelm Johannsen — He was a Danish botanist.

20. The location of a gene on a chromosome is its:

(a) Allele (b) Codon
(c) Locus (d) Promoter

Answer: (c) Locus — Plural: loci.

19.2 DNA AS THE HEREDITARY MATERIAL

21. Griffith worked on the bacterium:

(a) Escherichia coli (b) Bacillus subtilis
(c) Salmonella typhi (d) Streptococcus pneumoniae

Answer: (d) Streptococcus pneumoniae — It causes pneumonia in mice.

22. The S-type strain is virulent because it has a:

(a) Flagellum (b) Rough colony
(c) Polysaccharide coat (d) Protein coat

Answer: (c) Polysaccharide coat — The coat protects it from the host’s immune system.

23. Mice injected with heat-killed S-type bacteria alone:

(a) Died of pneumonia (b) Remained healthy
(c) Became carriers (d) Produced R-type bacteria

Answer: (b) Remained healthy — Dead bacteria cannot cause disease.

24. The change of R-type into S-type bacteria was called:

(a) Transformation (b) Transduction
(c) Translation (d) Translocation

Answer: (a) Transformation — Genetic material passed from dead S-type to live R-type.

25. "DNA is the fundamental unit of the transforming principle" was concluded by:

(a) Griffith (b) Avery, MacLeod and McCarty
(c) Hershey and Chase (d) Meselson and Stahl

Answer: (b) Avery, MacLeod and McCarty — In 1944.

26. Hershey and Chase used which organism in their main experiment?

(a) Bacteriophage (b) Pneumococcus
(c) Yeast (d) Pea plant

Answer: (a) Bacteriophage — A virus that attacks bacteria.

27. ^{35}S labelled which part of the bacteriophage?

(a) DNA (b) RNA
(c) Protein coat (d) Lipid envelope

Answer: (c) Protein coat — Sulphur is present in amino acids.

28. In the setup with ^{32}P-labelled phages, radioactivity was found in the:

(a) Medium only (b) Neither
(c) Protein coats (d) Bacterial cells

Answer: (d) Bacterial cells — The labelled DNA entered the bacteria.

29. The Hershey–Chase experiment was conducted in:

(a) 1928 (b) 1944
(c) 1952 (d) 1958

Answer: (c) 1952 — Griffith 1928, Avery 1944, Meselson–Stahl 1958.

19.3 DNA REPLICATION

30. DNA replication occurs during:

(a) S-phase of interphase (b) Prophase
(c) Metaphase (d) G2 phase

Answer: (a) S-phase of interphase — Before the cell divides.

31. In which model does each daughter DNA have one old and one new strand?

(a) Conservative (b) Dispersive
(c) Semi-conservative (d) None

Answer: (c) Semi-conservative — Each parent strand acts as a template.

32. In which model are strands mixtures of old and new nucleotides?

(a) Conservative (b) Dispersive
(c) Semi-conservative (d) Rolling

Answer: (b) Dispersive — The parent helix is broken into pieces.

33. The heavy isotope used by Meselson and Stahl was:

(a) ^{32}P (b) ^{14}C
(c) ^{35}S (d) ^{15}N

Answer: (d) ^{15}N — It entered the nitrogen bases of DNA.

34. Meselson and Stahl separated DNA by ultracentrifugation in a solution of:

(a) Cesium chloride (b) Sodium chloride
(c) Sucrose (d) Potassium chloride

Answer: (a) Cesium chloride — It forms a density gradient.

35. After the first replication in ^{14}N medium, the DNA showed:

(a) One heavy band (b) Two bands
(c) One light band (d) One intermediate band

Answer: (d) One intermediate band — Every DNA was a hybrid.

36. After the second replication, the DNA showed:

(a) One intermediate band
(b) One heavy and one light band
(c) One intermediate and one light band
(d) One light band

Answer: (c) One intermediate and one light band — This fits only the semi-conservative model.

37. The enzymes that break hydrogen bonds between base pairs are:

(a) Ligases (b) Helicases
(c) Primases (d) Gyrases

Answer: (b) Helicases — They are the unwinding enzymes.

38. The number of replication origin sites in a prokaryote is:

(a) Many (b) Thousands
(c) Two (d) One

Answer: (d) One — Eukaryotes have many.

39. The proteins that stop separated strands from re-joining are:

(a) Initiator proteins (b) Single-strand binding proteins
(c) Histones (d) Release factors

Answer: (b) Single-strand binding proteins — They bind to the single strands.

40. The primer is made of:

(a) 5–10 RNA nucleotides (b) 5–10 DNA nucleotides
(c) 100–200 RNA nucleotides (d) Amino acids

Answer: (a) 5–10 RNA nucleotides — It is attached by primase.

studysapiens.com

41. The main enzyme of DNA replication in prokaryotes is:

(a) DNA polymerase I (b) DNA polymerase II
(c) DNA polymerase III (d) Primase

Answer: (c) DNA polymerase III — It adds nucleotides and proofreads.

42. The enzyme that removes the primers and fills the gaps is:

(a) DNA polymerase III (b) Helicase
(c) DNA ligase (d) DNA polymerase I

Answer: (d) DNA polymerase I — It works at termination.

43. DNA polymerase II is mainly involved in:

(a) Repair of DNA damage (b) Primer synthesis
(c) Unwinding (d) Joining fragments

Answer: (a) Repair of DNA damage — During the lifetime of a cell.

44. New DNA strands are synthesised in the direction:

(a) 3′ to 5′ (b) Both directions
(c) 5′ to 3′ (d) No fixed direction

Answer: (c) 5′ to 3′ — Nucleotides are added only to the 3′ end.

45. Okazaki fragments are formed on the:

(a) Leading strand (b) Lagging strand
(c) Both strands (d) Template strand only

Answer: (b) Lagging strand — It is made discontinuously.

46. The region between the origin site and termination site is called a:

(a) Replication fork (b) Primer
(c) Replicon (d) Promoter

Answer: (c) Replicon — Also called a replication unit.

47. Okazaki fragments in eukaryotes are:

(a) 100–200 nucleotides (b) 1000–2000 nucleotides
(c) 5–10 nucleotides (d) 250 nucleotides

Answer: (a) 100–200 nucleotides — Those of prokaryotes are 1000–2000.

48. Eukaryotes have at least how many DNA polymerases?

(a) 3 (b) 20
(c) 5 (d) 14

Answer: (d) 14 — For example alpha, delta and epsilon.

49. The ability of DNA to be copied with accuracy is its:

(a) Variability (b) Stability
(c) Degeneracy (d) Universality

Answer: (b) Stability — It depends on proofreading.

studysapiens.com

19.4 GENE EXPRESSION AND GENETIC CODE

50. The term “Central Dogma” was used by:

(a) Nirenberg (b) Watson
(c) Khorana (d) Francis Crick

Answer: (d) Francis Crick — For the basic mechanism of gene expression.

51. The correct flow of information in the central dogma is:

(a) DNA → RNA → Protein (b) RNA → DNA → Protein
(c) Protein → RNA → DNA (d) DNA → Protein → RNA

Answer: (a) DNA → RNA → Protein — The flow is one-way.

52. The total number of codons is:

(a) 16 (b) 20
(c) 64 (d) 61

Answer: (c) 64 — 4 × 4 × 4.

53. A two-nucleotide code would give how many codons?

(a) 4 (b) 16
(c) 20 (d) 64

Answer: (b) 16 — 4 × 4, not enough for 20 amino acids.

54. The codon UUU specifies:

(a) Methionine (b) Arginine
(c) Glycine (d) Phenylalanine

Answer: (d) Phenylalanine — Shown by Nirenberg with poly-U mRNA.

55. Nirenberg and Leder determined the amino acids of:

(a) 17 codons (b) 61 codons
(c) 47 codons (d) 64 codons

Answer: (c) 47 codons — Khorana determined the remaining 17.

56. Which of these is a stop codon?

(a) AUG (b) UAA
(c) UUU (d) AGA

Answer: (b) UAA — The stop codons are UGA, UAG and UAA.

57. The start codon is:

(a) AUG (b) UAG
(c) UGA (d) GGC

Answer: (a) AUG — It specifies methionine.

studysapiens.com

58. The DNA codon TAC is transcribed into the mRNA codon:

(a) UAC (b) AUG
(c) ATG (d) TAC

Answer: (b) AUG — A pairs with U, and C with G.

59. AGA specifies arginine in bacteria as well as humans. This shows the code is:

(a) Universal (b) Degenerate
(c) Overlapping (d) Triplet

Answer: (a) Universal — The code is the same in almost all organisms.

60. A slightly different genetic code is found in the DNA of:

(a) Nucleus (b) Nucleolus
(c) Ribosomes (d) Mitochondria

Answer: (d) Mitochondria — An exception to universality.

61. Reading AUGAGCGCA only as AUG/AGC/GCA shows the code is:

(a) Degenerate (b) Universal
(c) Non-overlapping (d) Punctuated

Answer: (c) Non-overlapping — Each codon is read independently.

19.5 TRANSCRIPTION

62. The enzyme of transcription is:

(a) RNA polymerase (b) DNA polymerase
(c) Primase (d) Peptidyl transferase

Answer: (a) RNA polymerase — It needs no primer.

63. The DNA strand that is transcribed is the:

(a) Coding strand (b) Leading strand
(c) Sense strand (d) Template strand

Answer: (d) Template strand — It is also called the antisense strand.

64. The site on DNA where RNA polymerase binds is the:

(a) Operator (b) Promoter
(c) Replicon (d) Codon

Answer: (b) Promoter — It is not itself transcribed.

65. The –10 sequence of prokaryotes is:

(a) TTGACA (b) TATAAA
(c) TATAAT (d) GGCCAATCT

Answer: (c) TATAAT — TTGACA is the –35 sequence.

studysapiens.com

66. The TATA box of eukaryotes lies how many nucleotides before the start point?

(a) 25 (b) 10
(c) 35 (d) 70

Answer: (a) 25 — The CAAT box lies at –70.

67. The subunit of RNA polymerase that binds with template DNA is:

(a) α (b) σ
(c) β (d) β′

Answer: (d) β′ — β binds RNA nucleotides.

68. The core enzyme of RNA polymerase has how many subunits?

(a) Five (b) Four
(c) Three (d) Two

Answer: (b) Four — The σ subunit is removed after binding.

69. In eukaryotes, mRNA is synthesised by:

(a) RNA polymerase I (b) RNA polymerase III
(c) RNA polymerase II (d) Primase

Answer: (c) RNA polymerase II — I makes rRNA and III makes tRNA.

70. Transcription ends when the mRNA forms a:

(a) Poly-A tail (b) Primer
(c) 5′ cap (d) GC hairpin

Answer: (d) GC hairpin — It makes RNA polymerase pause.

71. The 5′ cap of mRNA is a:

(a) Poly-A chain (b) Methylated GTP
(c) Methionine (d) Histone

Answer: (b) Methylated GTP — It protects mRNA from nucleases and phosphatases.

72. The poly-A tail has about:

(a) 250 A nucleotides (b) 25 A nucleotides
(c) 2500 A nucleotides (d) 10 A nucleotides

Answer: (a) 250 A nucleotides — It is added at the 3′ end.

73. The non-coding regions of immature mRNA are called:

(a) Exons (b) Codons
(c) Introns (d) Operators

Answer: (c) Introns — They are removed in splicing.

studysapiens.com

19.6 TRANSLATION

74. The enzyme that attaches an amino acid to its tRNA is:

(a) Peptidyl transferase (b) Release factor
(c) RNA polymerase (d) Aminoacyl-tRNA synthetase

Answer: (d) Aminoacyl-tRNA synthetase — There are 20 kinds of it.

75. The first amino acid in prokaryotic translation is:

(a) N-formyl-methionine (b) Methionine
(c) Valine (d) Glycine

Answer: (a) N-formyl-methionine — It is a modified methionine.

76. The ribosomal site where the next aa-tRNA binds is the:

(a) P site (b) E site
(c) A site (d) Promoter

Answer: (c) A site — A = aminoacyl site.

77. Empty tRNAs leave the ribosome from the:

(a) A site (b) E site
(c) P site (d) 5′ cap

Answer: (b) E site — E = exit site.

78. The enzyme that forms peptide bonds is:

(a) Peptidyl transferase (b) Elongation factor
(c) Aminoacyl-tRNA synthetase (d) Ligase

Answer: (a) Peptidyl transferase — It joins the chain to the amino acid at the A site.

79. The movement of the ribosome by three nucleotides is called:

(a) Transformation (b) Translocation
(c) Transcription (d) Transposition

Answer: (b) Translocation — It occurs in the 5′–3′ direction.

80. A stop codon at the A site is recognized by the:

(a) Initiation factor (b) tRNA
(c) Release factor (d) Sigma subunit

Answer: (c) Release factor — It releases the polypeptide.

81. The ribosomes of prokaryotes are:

(a) 80S (b) 40S
(c) 60S (d) 70S

Answer: (d) 70S — Eukaryotes have 80S ribosomes.

studysapiens.com

19.7 REGULATING GENE EXPRESSION

82. Regulatory proteins that stop or reduce transcription are:

(a) Activators
(b) Release factors
(c) Inducers
(d) Repressors

Answer: (d) Repressors — This is negative regulation.

83. The piece of DNA near the promoter where regulatory proteins bind is the:

(a) Intron
(b) Operator
(c) Replicon
(d) Telomere

Answer: (b) Operator — Both repressors and activators bind here.

84. Activators regulate genes by:

(a) Blocking RNA polymerase
(b) Removing introns
(c) Helping RNA polymerase to bind to the promoter
(d) Breaking mRNA

Answer: (c) Helping RNA polymerase to bind to the promoter — This is positive regulation.

85. Production of more than one polypeptide from a single gene is due to:

(a) Alternate splicing
(b) Translocation
(c) Degeneracy
(d) Non-disjunction

Answer: (a) Alternate splicing — Introns or exons are removed in different ways.

86. The enzyme alcohol dehydrogenase is expressed in:

(a) Neurons
(b) Muscle cells
(c) Red blood cells
(d) Liver cells

Answer: (d) Liver cells — Neurons keep these genes turned OFF.

19.8 MUTATIONS

87. Agents that cause mutations are called:

(a) Mutagens
(b) Mutants
(c) Transposons
(d) Activators

Answer: (a) Mutagens — The affected organism is the mutant.

88. Exchange of a single nucleotide for another is a:

(a) Deletion
(b) Translocation
(c) Insertion
(d) Substitution

Answer: (d) Substitution — For example, A replaced by G.

studysapiens.com

89. X-rays and gamma rays damage DNA by:

(a) Joining pyrimidines
(b) Removing amino groups
(c) Breaking phosphodiester bonds through free radicals
(d) Adding hydrocarbon groups

Answer: (c) Breaking phosphodiester bonds through free radicals — They are ionizing radiation.

90. Mustard gas is an example of a:

(a) Physical mutagen (b) Mutagen chemical
(c) Transposon (d) Repressor

Answer: (b) Mutagen chemical — It is sometimes released into the environment.

91. A mutation that does not change the amino acid sequence is:

(a) Missense (b) Nonsense
(c) Silent (d) Deletion

Answer: (c) Silent — Example: GGC to GGU, both glycine.

92. A mutation that creates a premature stop codon is:

(a) Nonsense (b) Silent
(c) Missense (d) SNP

Answer: (a) Nonsense — Example: Duchenne muscular dystrophy.

93. Cystic fibrosis is commonly caused by the deletion of:

(a) One nucleotide (b) A chromosome
(c) A whole gene (d) Three nucleotides

Answer: (d) Three nucleotides — One amino acid is left out of the protein.

94. The gene for the β chain of haemoglobin is on chromosome:

(a) 12 (b) 11
(c) 21 (d) X

Answer: (b) 11 — The PAH gene is on chromosome 12.

95. In sickle cell anaemia, glutamic acid is replaced by:

(a) Valine (b) Glycine
(c) Tyrosine (d) Methionine

Answer: (a) Valine — CTC changes to CAC.

96. The enzyme deficient in phenylketonuria is:

(a) Alcohol dehydrogenase (b) Peptidyl transferase
(c) Phenylalanine hydroxylase (d) Tyrosinase

Answer: (c) Phenylalanine hydroxylase — It converts phenylalanine into tyrosine.

studysapiens.com

97. Tailoring drug treatment to a person's genetic makeup is:

(a) Gene therapy (b) Genetic counselling
(c) Cell therapy (d) Pharmacogenomics

Answer: (d) Pharmacogenomics — The patient's DNA is analysed first.

98. Mobile bits of DNA are called:

(a) Replicons (b) Transposons
(c) Introns (d) Primers

Answer: (b) Transposons — Also called jumping genes.

99. Failure of chromatids to separate in meiosis is:

(a) Non-disjunction (b) Translocation
(c) Inversion (d) Transformation

Answer: (a) Non-disjunction — A gamete gains or loses a chromosome.

100. Down syndrome is due to trisomy of chromosome:

(a) 13 (b) 18
(c) 21 (d) X

Answer: (c) 21 — The total number becomes 47.

101. The sex chromosomes in Klinefelter syndrome are:

(a) XO (b) XYY
(c) XXX (d) XXY

Answer: (d) XXY — The total is 44+XXY = 47.

102. The number of chromosomes in Turner syndrome is:

(a) 46 (b) 45
(c) 47 (d) 44

Answer: (b) 45 — 44+XO (monosomy).

103. Turner syndrome occurs roughly once in every:

(a) 750 births (b) 100 female births
(c) 5000 female births (d) 50,000 births

Answer: (c) 5000 female births — Down syndrome is about 1 in 750 children.

104. Estrogen replacement therapy is used in:

(a) Down syndrome (b) Phenylketonuria
(c) Klinefelter syndrome (d) Turner syndrome

Answer: (d) Turner syndrome — For secondary sexual characteristics.

105. A female with XXX sex chromosomes has:

(a) Turner syndrome (b) Triple X syndrome
(c) Klinefelter syndrome (d) Down syndrome

Answer: (b) Triple X syndrome — Such females are fertile.

studysapiens.com